

08
PHASE

AI-Era Analytics

Augment Intelligence. Automate Insights.
Predict. Personalize. Transform.



AI/ML
Models



Natural
Language
Processing



Intelligent
Automation



Predictive
Analytics



Personalized
Insights



Leverage
AI Models



Automate
Workflows



Discover
Deeper Insights



Make Smarter
Decisions



Drive Business
Transformation



The Future of Analytics is AI-Powered. Are You Ready?

Table of Contents

7 Units • 35 Topics • A Complete AI-Era Analytics Journey

| Unit 1: Advanced Prompt Engineering for Analytics.....12-27

1.1 The Analytical Persona: Crafting prompts that force the AI to act as a Senior Data Scientist.

1.2 Chain-of-Thought (CoT) Prompting: Forcing LLMs to explain their logic step-by-step to reduce math errors.

1.3 Few-Shot Learning: Providing examples of your schema and “Gold Standard” SQL to improve AI output.

1.4 Delimiters & Structured Output: Getting AI to return data in specific formats (JSON, Markdown, Python dictionaries).

1.5 Iterative Debugging: How to talk back to an AI when it gives you a hallucinated column name.

| Unit 2: AI-Assisted Development (SQL & Python)28-46

2.1 Copilot Mastery: Using GitHub Copilot, Cursor, or ChatGPT Canvas for real-time coding.

2.2 Refactoring & Optimization: Using AI to turn “spaghetti SQL” into clean, SARGable queries.

2.3 Auto-Documentation: Using AI to write dbt descriptions and Python docstrings.

2.4 The “Rubber Duck” Method: Using LLMs to explain complex legacy code you inherited.

2.5 Unit Testing with AI: Generating edge-case data to test if your Python functions break.

| Unit 3: AI-Powered Exploratory Data Analysis (EDA)47-67

3.1 PandasAI & Sketch: Natural language interfaces for Python DataFrames.

3.2 Automated Insight Discovery: Using AI to find anomalies or correlations you might have missed.

3.3 Qualitative Data at Scale: Using LLMs to categorize and sentiment-tag thousands of customer reviews in seconds.

3.4 Synthetic Data Generation: Creating “fake but realistic” datasets for privacy-safe testing.

3.5 Visualization Suggestions: Asking AI to choose the best chart based on the data distribution.

| Unit 4: The AI-Ready Semantic Layer (NL2SQL)68-92

- 4.1** What is a Semantic Layer?: Centralizing metrics so the AI knows exactly what “Revenue” means.
- 4.2** Metadata Management: Naming columns and tables so an LLM can understand them.
- 4.3** dbt Semantic Layer & Cube: Implementing a metrics layer that AI can query.
- 4.4** Vector Databases (Basics): How RAG (Retrieval-Augmented Generation) helps AI find the right documentation.
- 4.5** NL2SQL Interfaces: Setting up “Chat with your Data” for non-technical stakeholders.

| Unit 5: AutoML & Low-Code Predictive Analytics.....93-118

- 5.1** AutoML Overview: Using tools like PyCaret or Google Vertex AI to build models without writing the math.
- 5.2** Feature Importance: Using AI to tell you which variables actually drive your KPIs.
- 5.3** Automated Hyperparameter Tuning: Letting the machine find the best “settings” for your model.
- 5.4** Model Interpretation: Using SHAP or LIME (assisted by AI) to explain why a prediction was made.
- 5.5** From Analyst to “Citizen Data Scientist”: When to stop and hand it over to a Machine Learning Engineer.

| Unit 6: AI Ethics, Trust, & Governance.....119-146

- 6.1** The Hallucination Trap: Strategies for “Verifying the Bot” (The “Always Be Checking” rule).
- 6.2** Data Privacy: Understanding PII (Personally Identifiable Information) and safe LLM usage.
- 6.3** AI Bias Detection: Identifying if your AutoML model is discriminating against certain groups.
- 6.4** Governance Frameworks: Managing “Shadow AI” (unauthorized tool usage) in the workplace.
- 6.5** The “Human in the Loop”: Why AI provides the draft, but the Analyst provides the decision.

| Unit 7: Agentic Workflows (The Future of the Role)147-177

7.1 Introduction to Agents: AI that can use tools (SQL, Python, Web Search) in a sequence.

7.2 Building a Research Agent: Automating the collection of market data and competitor pricing.

7.3 Multi-Step Pipelines: Setting up an AI to “Run SQL → Clean in Python → Write Slack Summary.”

7.4 The Orchestration Layer: Basics of LangChain or AutoGen for data tasks.

7.5 Prompt Versioning: Treating your AI prompts like code (Git for Prompts).



Learnlytics

.handbook

LEARN SMART. ANALYZE BETTER. GROW FASTER.